

Algebra 1

Unit 13

Lesson H2 Homework

Name Answer Key

Date _____

Period _____

Use the functions $k(x) = 3x^2 + 4x - 5$ and $p(x) = 6 - 2x$ to determine each new function. Write your answer in standard form.

1. $(k + p)(x)$

$$\begin{array}{r} 3x^2 + 4x - 5 \\ + \quad \quad - 2x + 6 \\ \hline 3x^2 + 2x + 1 \end{array} \quad (k + p)(x) = 3x^2 + 2x + 1$$

2. $(k - p)(x)$

$$\begin{array}{r} (3x^2 + 4x - 5) - (6 - 2x) \\ 3x^2 + 4x - 5 - 6 + 2x \end{array} \quad (k - p)(x) = 3x^2 + 6x - 11$$

3. $(p - k)(x)$

$$\begin{array}{r} (6 - 2x) - (3x^2 + 4x - 5) \\ 6 - 2x - 3x^2 - 4x + 5 \end{array} \quad (p - k)(x) = -3x^2 - 6x + 11$$

4. $(k \cdot p)(x)$

$$\begin{array}{r} (3x^2 + 4x - 5)(6 - 2x) \\ 18x^2 + 24x - 30 - 6x^3 - 8x^2 + 10x \end{array} \quad (k \cdot p)(x) = -6x^3 + 10x^2 + 34x - 30$$

5. $\left(\frac{p}{p}\right)x$

$$\frac{p}{p} = 1 \quad \left(\frac{p}{p}\right)x = 1$$

6. Write the restriction on the domain for $\left(\frac{p}{p}\right)x$.

$$6 - 2x = 0$$

$$-2x = -6$$

$$x = 3$$

$$x \neq 3$$

Juan and Marina are renting a car for a cross country road trip. The car costs \$155 per day. The rental includes 100 miles for free and then charges \$0.25 per mile. The function $C(m) = 0.25(m - 100) + 155$ models the cost of the car, C , in terms of miles traveled, m . They determine gas will cost \$1.15 for each mile they travel. The company charges a \$40 fee for the first tank of gas. The function $G(m) = 1.15m + 40$ models the cost of gasoline for their vacation.

7. What is a reasonable domain for $C(m)$?

Answers may vary.

$m \geq 0$ (for max: approx. miles from east to west coast 2,500 – 3,500 miles 1 way)

8. What is a reasonable domain for $G(m)$?

Answers may vary.

$m \geq 0$ (for max: $\frac{3,500 \cdot 2}{\text{avg mpg}}$)

9. Determine the function that gives the total cost, $T(m)$, in dollars, based on the miles traveled.

$$T(m) = C(m) + G(m)$$

$$0.25(m - 100) + 155 + (1.15m + 40)$$

$$T(m) = 1.4m + 170$$

$$0.25m - 25 + 155 + 1.15m + 40$$

$$1.4m + 170$$

10. What is a reasonable domain for $T(m)$?

Answers may vary.

$m \geq 0$